FRUGAL-AI: A Cost-Aware Adaptive Framework for Reducing LLM Inference Cost

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Abstract

Large language model (LLM) inference now dominates the operating cost of AI applications, yet most production systems send every request to a single large model regardless of difficulty. Caching, routing and cascading each cut this cost, but they are usually studied one at a time and tuned to a quality target that is then missed on new data. We present FRUGAL-AI, a model-agnostic serving pipeline that combines a semantic cache, a difficulty router and a verifier-gated cascade, and calibrates all of their thresholds jointly against one quality target with a statistical margin. An online multiplier holds spend to a budget when traffic shifts. We evaluate on 885 queries from five benchmarks (multiple-choice question answering, competition mathematics, competitive programming, summarization and multi-turn dialogue) with a three-model Claude pool. At a target of 98% of the largest model's quality, on a stream with 30% repeated requests, FRUGAL-AI cuts cost by 50.8% ($\pm$ 5.2) and retains 98.5% of quality, against 20.5% for routing alone and 25.9% for caching alone. At matched quality on distinct queries it saves 21 points more than routing alone at 98% quality retained ($p = 0.003$) and 12 points more at 98.5% ($p = 0.04$); at 99% the two are statistically tied. We also report what did not work: prompt compression was never worth its quality loss, an online quality controller fed by a 2% audit sample gave no gain over the calibrated settings, a cascade without a router saved almost nothing, and per-workload quality floors could not be calibrated reliably for the full pipeline from 40 to 200 queries per workload. The online budget multiplier cut the share of windows over budget after a traffic shift from 86% to 9%, at a cost of about 4 points of quality. The code, the logged model outputs and the evaluation protocol are available at https://github.com/vijayakumargr/frugal-ai.

Keywords: large language models; inference cost optimization; model routing; model cascades; semantic caching; prompt compression; constrained optimization; green AI

1. Introduction

Most LLM spend is wasted on queries that a smaller model, a cached answer or a shorter prompt could serve equally well. Commercial APIs price per token, and the price gap between small and frontier models is often one to two orders of magnitude. Self-hosted deployments face the same gap in GPU hours and energy.

In practice, query difficulty is highly skewed. A large share of traffic in assistants, support bots and retrieval-augmented generation (RAG) systems is routine: greetings, FAQs, lookups and reformatting. Only a minority of requests needs multi-step reasoning. Sending all requests to the strongest model buys quality that most queries do not need.

Prior work mostly attacks this waste one technique at a time: caching, routing, cascading or prompt compression. Recent work adds online budget control to routing and statistical guarantees to cascades, but each still tunes one component. Two practical questions remain open. First, how much do the techniques add to each other when they are tuned together against a single quality target? Second, does a system tuned to a target on calibration data actually meet it on new traffic? Reported savings are hard to compare when methods land at different quality levels.

Research questions.

1. RQ1: How much cost can a combined pipeline save, at a fixed quality target, compared with each technique alone?

2. RQ2: Can an online controller keep quality above a target and spend under a budget as traffic shifts?
3. RQ3: Which components contribute most, and do their gains add up or overlap?

Contributions.

- A serving pipeline, FRUGAL-AI, that composes a semantic cache, a difficulty router and a verifier-gated cascade, with all thresholds calibrated jointly against one quality target, a margin for sampling error and a share of the tolerance reserved for wrong cache hits.
- A formulation of LLM serving as constrained expected-cost minimization, with a primal-dual controller whose budget multiplier we show to hold spend under a traffic shift.
- A controlled evaluation on logged outputs of three models, in which every method is scored on identical answers and compared both at a common target and at matched achieved quality.
- Negative results reported in full: prompt compression, an audit-driven online quality controller and per-workload quality floors did not pay off in this setting, and a cascade without a router saved almost nothing.
- An open-source implementation and the logged model outputs, so that all tables can be reproduced without new model calls.

Section 2 reviews related work. Section 3 states the problem, Section 4 describes the framework and Section 5 its algorithms. Sections 6 and 7 present the experiments, and Sections 8 and 9 discuss limitations and conclude.

2. Related Work

Existing cost-reduction methods each tune one source of waste. FRUGAL-AI calibrates the cache, router and cascade together, and our contribution is the measurement of what that joint calibration gains and where it fails, not a claim to the first budget-controlled router.

Model cascades and routing. FrugalGPT [1] showed that a cascade of LLM APIs with a learned scorer can match a strong model at lower cost. Hybrid LLM [2] and RouteLLM [3] train routers that choose between a small and a large model per query, using preference data. AutoMix [4] lets a small model self-verify and escalate. These methods use fixed thresholds and do not adapt to budget or traffic drift.

Online and guaranteed routing (2025–2026). Several recent works close the gap on adaptation; a survey is given in [37]. ParetoBandit [30] and WISERouter [31] treat routing as a constrained contextual bandit with online budget pacing, GAR [32] uses an online primal-dual router with accuracy floors, and Panda et al. [33] give a bandit router under a budget. Cascade routing [34] unifies routing and cascading in one decision rule. Conformal Cascade [35] and C3PO [36] give distribution-free guarantees for cascade deferral and cascade cost. These works control one component, the router or the cascade; none calibrates a semantic cache together with them, and none reports whether the calibrated quality target is met on held-out traffic for a full pipeline. We did not implement them as baselines (Section 8).

Caching. GPTCache [5] reuses answers for semantically similar queries using embedding search. Key-value cache reuse and paged attention [6] reduce serving cost inside the model server but do not avoid model calls.

Prompt compression. Selective Context [7] and LLMLingua [8], [9] drop low-information tokens using a small language model, reducing input tokens with modest quality loss.

Model compression and efficient decoding. Distillation [10], quantization such as GPTQ [11] and AWQ [12], and speculative decoding [13] make each call cheaper. They are orthogonal to FRUGAL-AI and can be applied to any model in its pool.

Constrained online learning. Bandits with knapsacks [14] and online convex optimization with long-term constraints [15], [16] give the tools we use to keep spend and quality within limits over time.

Green AI. Strubell et al. [17] and Schwartz et al. [18] argue for reporting compute and energy cost alongside accuracy, which motivates our evaluation protocol.

Approach	Cache	Compress prompt	Route per query	Cascade	Budget control online	Quality target calibrated
FrugalGPT [1]	yes	partial	no	yes	no	no
Hybrid LLM [2]	no	no	yes	no	no	no
RouteLLM [3]	no	no	yes	no	no	no
AutoMix [4]	no	no	no	yes	no	no
GPTCache [5]	yes	no	no	no	no	no
LLMLingua [8]	no	yes	no	no	no	no
ParetoBandit [30], WISERouter [31]	no	no	yes	no	yes	no
Cascade routing [34]	no	no	yes	yes	no	no
Conformal Cascade [35]	no	no	no	yes	no	yes, for the cascade
FRUGAL-AI (ours)	yes	optional, unused	yes	yes	yes	yes, jointly, with a margin; met in 70% of seeds

3. Problem Formulation

We cast LLM serving as choosing, per query, the cheapest action whose expected quality keeps the long-run average above a target.

Let M = {m_1, …, m_K} be a pool of models ordered by per-token price. For a query x, model m_k incurs cost $c_k(x) = p_k^{\text{in}} \cdot n_{\text{in}}(x) + p_k^{\text{out}} \cdot n_{\text{out}}(x)$, where p_k are the input and output prices per token and n are token counts. A quality function Q(y, x) $\in$ [0, 1] scores the answer y, using task metrics or an LLM judge.

A serving policy π maps each query to an action a $\in$ A: return a cached answer, compress the prompt, call one model, or run a cascade of models. Let $C_\pi(x)$ and $Q_\pi(x)$ be the realized cost and quality. Given a quality target q^* and a per-request budget B, we solve:

$$\min_{\pi} \ \mathbb{E}_{x\sim\mathcal{D}}[C_\pi(x)] \quad \text{s.t.} \quad \mathbb{E}_{x\sim\mathcal{D}}[Q_\pi(x)] \geq q^*, \qquad \mathbb{E}_{x\sim\mathcal{D}}[C_\pi(x)] \leq B$$

In practice q^* is set relative to a reference policy that always calls the strongest model m_K, for example q^* = $(1 - \varepsilon) \cdot \mathbb{E}[Q_K(x)]$ with tolerance ε such as 0.01 or 0.02.

The Lagrangian relaxation with multiplier $\lambda \geq 0$ gives a per-query rule: choose the action that minimizes predicted cost minus λ times predicted quality.

$$a^*(x) = \arg\min_{a\in\mathcal{A}} \ \hat{c}_a(x) - \lambda\, \hat{q}_a(x)$$

Here $\hat{c}$ and $\hat{q}$ are learned estimates. A large λ favors quality and a small λ favors savings. Section 5 shows how λ is adjusted online so that both constraints hold over time.

For a cascade that tries models $k_1 < k_2 < \cdots < k_J$ in order and stops at stage j when a verifier accepts, the expected cost is:

$$\mathbb{E}[C_{\text{cascade}}(x)] = \sum_{j=1}^{J} \left(c_{k_j}(x) + v_{k_j}(x)\right) \cdot \prod_{i<j} \left(1 - \alpha_{k_i}(x)\right)$$

where v is the verifier cost and $\alpha_k(x)$ is the probability that the verifier accepts the answer of m_k. Note that the product multiplies both terms in the sum. A cascade pays off only when cheap stages are accepted often enough to offset the cost of failed attempts.

4. The FRUGAL-AI Framework

FRUGAL-AI is a pipeline in front of a model pool: a semantic cache, an optional prompt compressor, a difficulty router and a verifier-gated cascade. Each stage can end the request early, and one calibration step sets all thresholds against a single quality target. In the configuration evaluated as FRUGAL-AI in Section 7 the compressor is switched off.

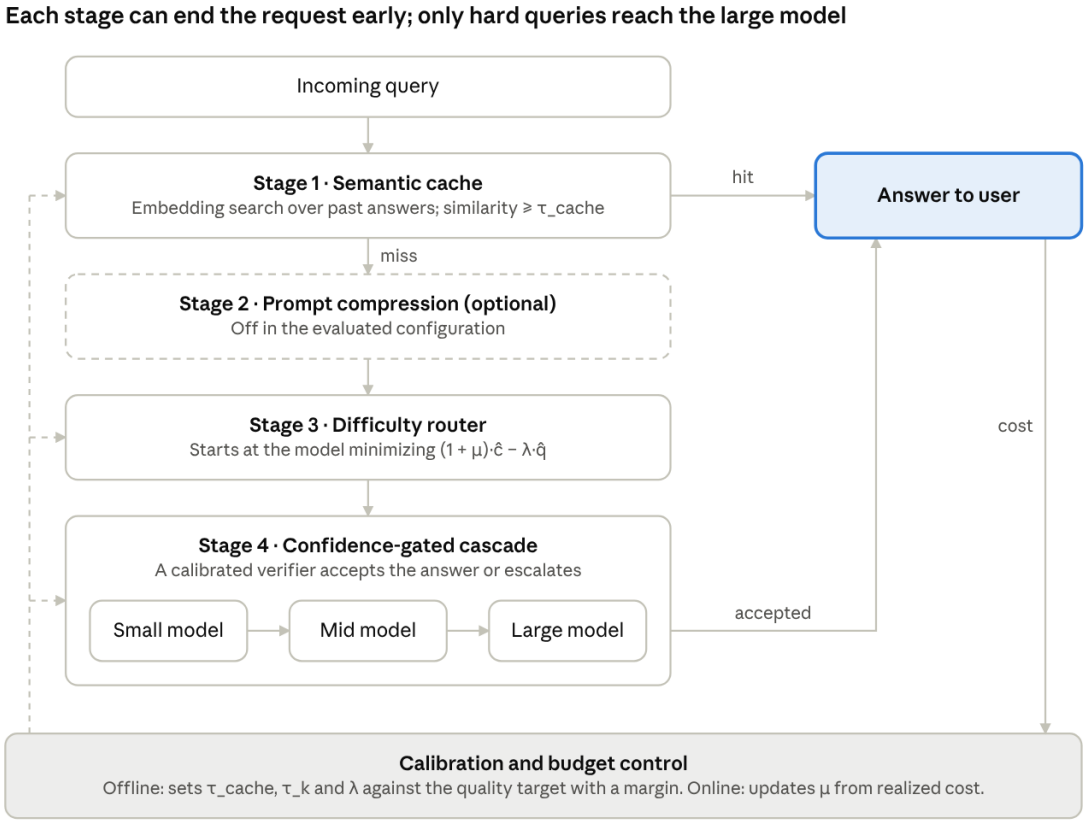


Figure 1. A cache hit or an accepted cascade answer ends the request. The thresholds are calibrated offline against the quality target, and the budget multiplier μ is updated online from realized cost. Prompt compression (dashed) is optional and off in the evaluated configuration.

4.1 Stage 1: Semantic cache

Each query is embedded with a small sentence encoder and searched in a vector index of past (query, answer) pairs. If the nearest neighbour has cosine similarity above τ_{cache} and the cached answer passed verification, it is returned at near-zero cost. To limit stale or wrong reuse, entries carry a time-to-live, and the cache is keyed on the query plus a hash of any retrieved context. The threshold τ_{cache} is the lowest value whose conformal bound on the false-hit rate, measured on a calibration replay stream, stays within the cache's share of the quality tolerance. We reserve a quarter of the tolerance ε for wrong cache hits, so at $\varepsilon = 0.02$ the bound is 0.5%.

4.2 Stage 2: Task-aware prompt compression

Long prompts in RAG and multi-turn chat dominate input cost. A small model scores each context sentence for relevance to the query, using token-level information in the style of [7], [8]. Sentences below a learned keep-ratio r are dropped, while instructions, the query and code blocks are always kept. The ratio r is chosen per task to bound the predicted quality loss.

In our implementation the compressor keeps the half of the document sentences most similar to the query. The compressed prompt is one more action the router may choose. On the summarization workload it lowered the judged score from about 0.80 to about 0.70, so the calibration never selected it at the targets we tested; the stage is therefore off by default and reported as an ablation.

4.3 Stage 3: Difficulty-aware router

The router predicts, for each model m_k, the probability $\hat{q}_k(x)$ that m_k's answer meets the quality bar. It is a light classifier, such as a fine-tuned small encoder or gradient-boosted trees on embeddings plus features like query length, domain tag and presence of code or math. Training labels come from running all pool models on a training set and scoring their answers. The router then applies the Lagrangian rule from Section 3 to pick either a single model or the starting point of a cascade.

4.4 Stage 4: Confidence-gated cascade

If the router starts at a cheaper model, a verifier decides whether to accept its answer or escalate to the next model. The verifier is one call to the small model that rates the answer's correctness from 0 to 100; the API exposes no token log-probabilities. A logistic model calibrates this rating, together with the router's prior, into a probability that the answer is good. The acceptance threshold τ_k is set with conformal risk control [26], which bounds the expected quality lost to wrongly accepted answers by a level α.

4.5 Budget controller

Offline calibration. The controller first sets the cache threshold, the cascade risk level α (and with it the thresholds τ_k) and the quality multiplier λ on the calibration split. It lists every candidate setting with its mean cost and quality, keeps the cost-quality front, and walks that front from the safest setting towards cheaper ones. A setting is accepted while its mean quality exceeds the target by one standard error of the paired difference to the reference model; the walk stops at the first setting that fails. The routing stages are calibrated to three quarters of the tolerance ε, and the cache to the remaining quarter. The same rule is applied to every tuned baseline.

Online control. While serving, the controller updates the budget multiplier μ with the primal-dual rule in Section 5; a positive μ also lowers τ_k, so more cheap answers are accepted under budget pressure. An online quality multiplier, driven by a random audit sample that also calls the reference model (2% of traffic, with the audit calls counted in the cost), is available as an option. It gave no gain in our experiments and is off by default.

4.6 Design properties

- **Model-agnostic:** works with API models and self-hosted models, and with compressed or quantized models in the pool.
- **Composable:** each stage can be switched off, which enables the ablations in Section 7.
- **Low overhead:** all control components are small models that run on CPU or a shared GPU.

5. Algorithms

This section gives the two multiplier updates used online, the calibration and serving algorithms, and their relation to existing theory. In our experiments only the budget multiplier was effective; Section 7.2 reports why the quality multiplier was not.

5.1 Online primal-dual control

We keep two multipliers: λ_t for the quality constraint and μ_t for the budget constraint. At each request t the router picks the action that minimizes the per-query Lagrangian:

$$a_t = \arg\min_{a \in \mathcal{A}} (1 + \mu_t)\, \hat{c}_a(x_t) - \lambda_t\, \hat{q}_a(x_t)$$

After serving, the controller updates the multipliers by projected subgradient steps with step size η:

$$\lambda_{t+1} = \left[\lambda_t + \eta\,(q^* - \tilde{Q}_t)\right]_+, \qquad \mu_{t+1} = \left[\mu_t + \eta\,(C_t - B)\right]_+$$

In the implementation, cost is measured in units of the reference model's mean cost, and a positive μ also divides the acceptance thresholds τ_k by $1 + \mu$. Quality is observed only on audited requests, a random share ρ of traffic on which the reference model also answers. On those requests the quality step uses the gap between $(1 - \varepsilon)$ times the reference answer's quality and the served answer's quality, scaled by $1/\rho$ so that the update is unbiased; on all other requests λ is unchanged. The quality update is optional and off in the default configuration.

5.2 Algorithms 1 and 2: calibration and serving

Algorithm 1 is run once on the calibration split and fixes the cache threshold, the cascade thresholds and the quality multiplier. Algorithm 2 then serves requests with those settings and, when a budget is given, updates the budget multiplier.

```
# Algorithm 1: offline calibration (calibration split only, pseudocode)
def calibrate(settings, ref_quality, eps, z=1.0, cache_share=0.25):
    # Cache: lowest threshold whose conformal bound on wrong hits fits its share.
    tau_cache = min(t for t in cache_grid
                    if (false_hits(t) + 1) / (n_requests + 1) <= cache_share * eps)
    # Routing and cascade: the rest of the tolerance, per query.
    target = (1 - (1 - cache_share) * eps) * ref_quality
    front = pareto_front(settings)        # by mean cost and mean quality
    chosen = front.safest()               # always-large
    for s in front.from_safest_to_cheapest():
        d = s.quality - target            # paired difference per query
        if mean(d) - z * stderr(d) < 0:
            break                         # the first failure ends the walk
        chosen = s
    return tau_cache, chosen.tau, chosen.lam
```

```
# Algorithm 2: serving loop, default configuration (pseudocode)
mu = 0.0
for x in stream:
    hit = cache.lookup(embed(x), tau_cache)
    if hit is not None:
        y, cost = hit.answer, 0.0
    else:
        q_hat, c_hat = router.predict(x)  # expected cascade quality and cost per start model
        k = argmin((1 + mu) * c_hat - lam * q_hat)
        cost = 0.0
        while True:
            y = models[k].generate(x)
            cost += price(k, x, y)
            if k == K - 1:
                break                     # the largest model's answer is final
            cost += verifier_price
            if verifier.score(x, y) >= tau[k] / (1 + mu):
                break                     # accepted; budget pressure relaxes acceptance
            k += 1                        # escalate
        cache.insert(x, y)
    if budget is not None:
        mu = max(0.0, mu + eta_mu * (cost / ref_cost - budget))
    yield y
# Options, off by default: a compressed prompt as an extra action for the router,
# and the audit-driven update of lam from Section 5.1.
```

5.3 Relation to existing theory

We do not claim a new theoretical result. The multiplier updates of Section 5.1 follow the dual-subgradient scheme analysed for online convex optimization with long-term constraints [15] and with stochastic constraints [16]; for i.i.d. requests and a strictly feasible policy, the latter gives expected regret and constraint violation of order $\sqrt{T}$. Our setting departs from the assumptions of those results in three ways: the action set is discrete and chosen through learned cost and quality estimates, quality is observed only on a small audit sample, and traffic may shift. The bounds therefore do not transfer directly, and we make no formal

claim for the online controller. Empirically (Section 7.2) the budget multiplier held spend under a shift, while the quality multiplier, with about 10 audit labels per 500 requests, did not track quality.

The offline calibration gives a different kind of assurance. For each cascade threshold taken alone, conformal risk control [26] bounds the expected quality lost to wrongly accepted answers by α under exchangeability, and the same conformal correction bounds the cache's false-hit rate. The joint choice of the risk level and the multiplier in Algorithm 1, with its one-standard-error margin and its stop at the first failing setting, is a heuristic for which we give no guarantee. Section 7 therefore reports how often the target is met on held-out data.

5.4 Complexity and overhead

Per request, the cache lookup costs O(log N) with an approximate nearest-neighbour index over N entries. The router and cost model cost one small forward pass each. The cascade adds at most $K - 1$ extra model calls in the worst case, and the controller update is O(1). We report measured overhead in Section 7.

6. Experimental Setup

We evaluate on five public workloads with a three-tier model pool and compare against six baselines at matched quality targets.

6.1 Workloads

Workload	n	Task	Quality metric	Why included
MMLU-Pro [27]	400	Multiple-choice QA with up to ten options	Accuracy	Separates the three models
Codeforces problems from CodeContests [28]	165	Competitive programming in Python	All hidden tests pass (up to 15 tests, 10 s each)	Verifiable correctness; hard for small models
AIME 2022–2024 [29]	90	Competition mathematics	Exact-match integer answer	Multi-step reasoning
CNN/DailyMail [24]	150	Long-document summarization	LLM-judge score (1–10, scaled to 0–1)	Long prompts test compression
MT-Bench [25]	80	Multi-turn open dialogue	LLM-judge score (1–10, scaled to 0–1)	Open-ended, assistant-like traffic

The total is 885 queries. A 100-query pilot on MMLU [21], GSM8K [22] and HumanEval [23] showed that the small model reached about 98% of the large model's quality there, so those benchmarks were replaced with the harder ones above. To test caching, repeated and paraphrased requests are added to the stream (Section 6.5). The Codeforces problems date from 2021 and earlier, so the models may have seen them in training. A real, consented production log would strengthen the paper considerably.

6.2 Model pool

The pool has three tiers from one model family: Claude Haiku 5.5 (small), Claude Sonnet 5.5 (mid) and Claude Opus 5.5 (large, the reference m_K), all called through the Anthropic API at the same effort setting. Costs use list prices as of October 2026, in US dollars per million input / output tokens: 0.10 / 0.50, 2.00 / 10.00 and 4.00 / 20.00. A single-family pool is a limitation; an open-weight tier would test generality.

Cost accounting. Calls were made through the Claude Code command line in non-interactive mode, with an empty system prompt and no tools, not through the metered API. Cost is therefore computed as logged tokens times list prices. The command line adds about 550 input tokens to every call; this overhead was measured once per model and subtracted. The whole collection came to 7,885 calls and 47.45 US dollars at list prices, with no refusals, truncated answers, timeouts or judge parse failures.

6.3 Baselines

1. **Always-large:** every query goes to m_K (quality reference).
2. **Always-small:** every query goes to the cheapest model (cost floor).
3. **Cache-only:** semantic cache in front of m_K, as in [5].
4. **Compression-only:** LLMLingua-style compression [8] in front of m_K.
5. **Router-only:** a RouteLLM-style binary router [3].
6. **Cascade-only:** FrugalGPT/AutoMix-style cascade with fixed thresholds [1], [4].

Router-only, cascade-only and cache-only are tuned with the same calibration rule and margin as FRUGAL-AI. We also report two variants that calibrate one setting per workload, so that each workload must meet the target on its own: Router-only with per-workload floors and FRUGAL-AI with per-workload floors. The 2026 systems discussed in Section 2 are not implemented.

6.4 Metrics

- **Cost:** US dollars per 1,000 requests, and cost reduction relative to always-large.
- **Quality:** task metric, and quality retained as a percentage of always-large.
- **Pareto frontier:** cost versus quality as q^* sweeps from 0.90 to 1.00 of the reference.
- **Budget adherence:** share of time windows where spend exceeded B, and worst overshoot.
- **Latency:** 95th-percentile end-to-end latency.
- **Energy (optional):** estimated kWh and CO2e for self-hosted models, following [17], [18].

Two further measures make comparisons fair. Seeds meeting target is the share of seeds in which a method's quality on the test stream reached q^*. Cost reduction at matched quality reads each method's saving at a fixed quality retained (97%, 98%, 98.5% and 99%) by interpolating along its cost-quality front on the test split, with always-large and always-small as end points; a point between two settings can be realized by randomizing between them.

6.5 Protocol

Routers and verifiers are trained, and all thresholds calibrated, on a 50% split that does not overlap the test split, with 5-fold cross-fitting. All model outputs are logged once and replayed, so every method is scored on identical outputs. Every result is the mean over 10 seeds with 95% confidence intervals; a seed redraws the split, the request stream and the audit sample, not the model outputs. Significance uses paired t-tests across seeds. The main target is $q^* = 0.98 \times$ always-large ($\varepsilon = 0.02$); we also report $\varepsilon = 0.01$.

Disclosure. A first evaluation tuned every method by taking the cheapest setting that met the target on the calibration split, at $\varepsilon = 0.01$ and with 5 seeds. FRUGAL-AI then retained 97.7% against a 99% target, and no tuned method reached the target on the test split. We replaced that rule with the margin-based calibration of Section 4.5, moved the main target to 98% and raised the number of seeds to 10. The rule was therefore chosen after seeing test results from the first evaluation; the first results are kept with the released code.

The cache is tested on a replay stream in which 30% of requests are repeats, half of them paraphrased by the small model. This share is an assumption and is varied from 0% to 50%. To test drift (RQ2), the share of easy queries falls from 80% to 50% mid-stream.

7. Results

At a 98% quality target FRUGAL-AI saves about half the cost of always calling the large model, roughly twice the saving of routing or caching alone at similar quality; at a 99% target its advantage over routing shrinks to within the noise. On all 885 queries the three models score 0.824 (small), 0.856 (mid) and 0.891 (large), at 1.53, 10.71 and 22.84 US dollars per 1,000 requests. All intervals are 95% confidence intervals over 10 seeds. Each test split holds about 440 queries, so differences under about 1 point of quality are not resolved.

7.1 Main results (RQ1)

Table 2. Replay stream (30% repeated requests), target $q^* = 0.98 \times$ always-large. Mean over 10 seeds ($\pm$ 95% CI).

Method	Cost (USD / 1k requests)	Cost reduction (%)	Quality retained (%)	p95 latency (s)	Seeds meeting target (%)
Always-large	22.81 ± 1.84	0	100	40.6 ± 3.0	100
Always-small	1.52 ± 0.13	93.3 ± 0.5	92.4 ± 1.6	52.3 ± 7.3	0
Cache-only	16.88 ± 1.29	25.9 ± 2.8	99.3 ± 0.4	34.1 ± 3.1	100
Compression-only	22.35 ± 1.88	2.1 ± 0.4	97.6 ± 0.4	40.6 ± 3.0	40
Router-only	18.19 ± 2.17	20.5 ± 4.4	98.7 ± 1.2	37.3 ± 3.5	80
Cascade-only	21.41 ± 3.29	6.9 ± 9.8	98.2 ± 0.9	134.8 ± 17.7	60
FRUGAL-AI	11.20 ± 1.48	50.8 ± 5.2	98.5 ± 0.7	57.6 ± 12.7	70
Router-only, per-workload floors	18.59 ± 2.67	19.0 ± 7.5	99.4 ± 0.5	38.5 ± 5.1	100
FRUGAL-AI, per-workload floors	13.15 ± 1.57	42.2 ± 5.8	97.7 ± 0.8	65.6 ± 10.3	30

FRUGAL-AI costs 5.67 US dollars per 1,000 requests less than cache-only, 6.99 less than router-only and 10.20 less than cascade-only (paired t-tests across seeds, $p \leq 0.0001$ each). Its mean quality retained, 98.5%, is above the target and within 0.2 points of router-only, but it meets the target in 7 of 10 seeds against 8 of 10 for router-only. The price is latency: verifier calls and escalations raise p95 latency from 40.6 s to 57.6 s. Cascade-only saves little and more than triples p95 latency, because every answer from a cheaper model also pays for a verifier call.

Table 3. Distinct queries only (no repeats, so the cache can only do harm), target 98%.

Method	Cost reduction (%)	Quality retained (%)	p95 latency (s)	Seeds meeting target (%)
Cache-only	0.9 ± 0.7	99.3 ± 0.5	42.5 ± 4.6	90
Router-only	21.1 ± 4.8	98.9 ± 0.6	40.1 ± 4.4	90
Cascade-only	6.7 ± 8.9	98.0 ± 0.5	142.6 ± 15.2	50
FRUGAL-AI	33.8 ± 7.0	98.1 ± 0.5	86.7 ± 21.1	50
Router-only, per-workload floors	18.9 ± 6.9	99.1 ± 0.4	41.5 ± 5.4	100
FRUGAL-AI, per-workload floors	22.8 ± 7.1	97.6 ± 0.6	91.1 ± 17.3	50

Without repeats FRUGAL-AI still saves 12.7 points more than router-only, but it sits 0.8 points lower in quality and meets the target in only half the seeds, so this comparison is not at equal quality. Table 5 and Figure 2 make the comparison at matched quality.

Table 4. Stricter target, $q^* = 0.99 \times$ always-large. Each cell is cost reduction (%) / quality retained (%) / seeds meeting target (%).

Method	Replay stream	Distinct queries
Cache-only	22.0 ± 3.1 / 99.8 / 100	0.2 ± 0.3 / 99.8 / 90
Router-only	15.3 ± 4.2 / 99.6 / 90	15.5 ± 3.8 / 99.4 / 90
Cascade-only	−22.0 ± 10.1 / 99.3 / 80	−21.8 ± 10.2 / 99.1 / 60
FRUGAL-AI	35.4 ± 5.9 / 99.1 / 50	17.1 ± 6.6 / 98.9 / 40

At the 99% target FRUGAL-AI's saving on the replay stream comes mostly from the cache: on distinct queries it saves about the same as router-only and meets the target less often. With about 440 calibration

queries, a 1% tolerance is close to the sampling error of the quality estimate itself, and the simpler method, with one threshold to calibrate, is the more reliable one.

Table 5. Cost reduction (%) at matched quality retained, distinct queries.

Quality retained	FRUGAL-AI	Router-only	Cascade-only	FRUGAL-AI minus Router-only	p-value
97%	69.5 ± 4.2	41.0 ± 5.7	31.1 ± 8.2	+28.5 ± 7.8	< 0.0001
98%	53.5 ± 9.1	32.4 ± 4.8	12.0 ± 5.9	+21.1 ± 11.7	0.003
98.5%	41.1 ± 8.7	28.8 ± 4.6	7.4 ± 3.9	+12.3 ± 11.7	0.04
99%	28.9 ± 5.8	24.7 ± 4.7	4.8 ± 2.7	+4.2 ± 9.8	0.36

The advantage of the combined pipeline over routing alone is large at 97% to 98% quality retained, still significant at 98.5%, and gone at 99%.

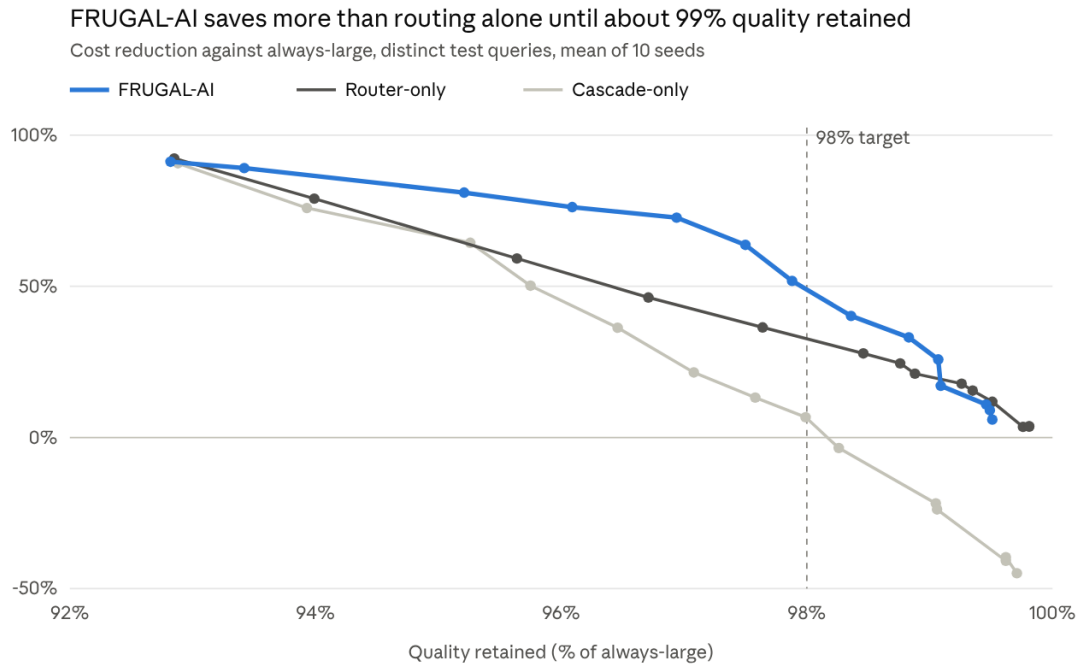


Figure 2. Cost-quality frontier on distinct queries. Each point is one quality tolerance, from 10% down to 0%. Below zero a method costs more than always calling the large model.

Table 6. By workload at the 98% target, distinct queries. Each cell is cost reduction (%) / quality retained (%).

Workload	Router-only	FRUGAL-AI	Router-only, per-workload floors	FRUGAL-AI, per-workload floors
AIME	90.2 / 95.8	64.4 / 97.6	26.5 / 98.0	71.0 / 97.1
CNN/DailyMail	5.3 / 99.5	34.5 / 98.1	23.9 / 98.1	19.6 / 98.9
Codeforces	1.5 / 99.9	14.4 / 96.6	11.7 / 98.6	8.9 / 96.6
MMLU-Pro	32.4 / 99.2	11.7 / 99.2	24.4 / 100.1	30.2 / 97.6
MT-Bench	31.8 / 98.1	78.8 / 96.1	26.9 / 98.8	25.8 / 98.3

A target on average quality lets individual workloads fall below it: under FRUGAL-AI, Codeforces retains 96.6% and MT-Bench 96.1%, and under router-only AIME retains 95.8%. Calibrating one setting per

workload repairs this for the router, which then meets the overall target in every seed. It does not work for FRUGAL-AI: with 40 to 200 calibration queries per workload and many more settings to choose from, its per-workload calibration overfits, and the overall target is met in only 3 of 10 seeds on the replay stream.

7.2 Budget adherence under drift (RQ2)

The budget multiplier held spend under a traffic shift; the online quality multiplier did not hold quality. When the share of easy queries falls from 80% to 50%, FRUGAL-AI with fixed calibrated settings goes from 98.8% to 95.6% quality retained and from 32.8% to 21.9% cost reduction. Adding the online quality controller changes almost nothing: 98.7% to 95.4%, with 42.0% of windows more than 1 point below the target against 40.5% without it. A 2% audit rate gives about 10 quality labels per 500-request window, too few to move the multiplier in time. The fixed cascade baseline degrades further, from 97.9% to 93.7% quality retained and from a 6.7% saving to a 12.3% cost increase. No method tested keeps quality at the target after the shift.

Table 7. Budget control after the shift. The budget is 15% above the pre-shift spend of the fixed policy.

Policy	Windows over budget (%)	Worst overshoot (%)	Quality retained after shift (%)
Fixed thresholds	86.0 ± 21.1	68.5 ± 27.6	95.6 ± 1.5
Online budget control	9.0 ± 10.4	−2.0 ± 5.0	91.5 ± 2.3

The budget multiplier keeps spend under the budget by accepting more answers from cheaper models, which costs about 4 points of quality. It is also conservative: because per-request cost is heavy-tailed, the multiplier stays positive even before the shift, and spend settles well below the budget instead of at it (Figure 3).

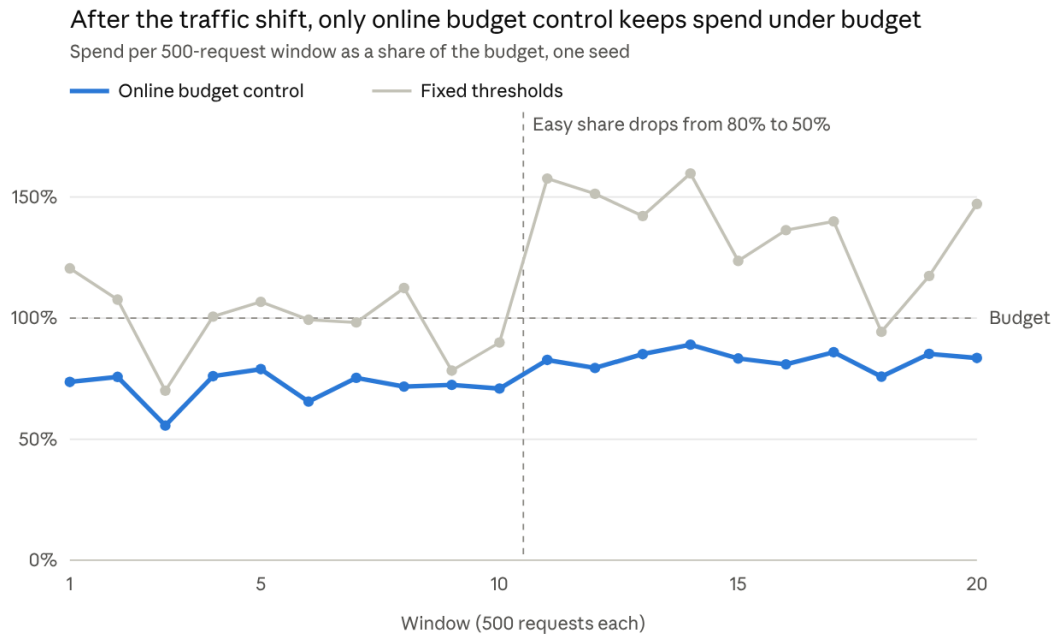


Figure 3. Spend per window for one seed. The share of easy queries drops after window 10.

7.3 Ablation (RQ3)

Table 8. Ablation on the replay stream at the 98% target. Each variant is recalibrated with the same rule.

Configuration	Cost reduction (%)	Quality retained (%)	Seeds meeting target (%)	Change in cost reduction (points)
Full FRUGAL-AI	50.8 ± 5.2	98.5 ± 0.7	70	0
without cache	40.4 ± 8.5	98.7 ± 0.8	90	−10.4
without router (cascade from smallest)	12.8 ± 7.5	98.5 ± 0.7	80	−38.0
without cascade (three-way router)	36.0 ± 8.4	98.6 ± 0.6	80	−14.8
with prompt compression	50.8 ± 5.2	98.5 ± 0.7	70	0.0
with online quality controller	48.8 ± 4.9	98.5 ± 0.7	70	−2.0

All three core stages contribute at similar quality. The router matters most: a cascade that always starts at the small model saves only 12.8%. The verifier-gated cascade adds 14.8 points on top of a three-way router, because it lets the router start low on queries it is unsure about and still recover. The cache adds 10.4 points on this stream, less than its 28% hit rate, since many repeated requests would have been served cheaply anyway. Allowing prompt compression changes nothing, because the calibration never selects it. The online quality controller costs 2 points, the price of its audit calls, and returns no quality.

7.4 Sensitivity and overhead

Results move mainly with the share of repeated requests and the cache threshold.

Table 9. Sensitivity of FRUGAL-AI on the replay stream, 98% target.

Setting	Cost reduction (%)	Quality retained (%)	Cache hit rate (%)	False-hit rate (%)
$\tau_{\text{cache}} = 0.80$	52.6 ± 4.9	97.6 ± 0.9	30.9	1.53
$\tau_{\text{cache}} = 0.85$	51.3 ± 5.2	98.5 ± 0.9	29.8	0.62
$\tau_{\text{cache}} = 0.90$	50.6 ± 5.4	98.7 ± 0.9	28.5	0.36
$\tau_{\text{cache}} = 0.95$	50.1 ± 5.4	99.0 ± 0.8	26.8	0.16
Repeat share 0%	33.4 ± 7.2	98.2 ± 0.5	0.7	0.70
Repeat share 15%	43.2 ± 6.5	98.3 ± 0.6	14.4	0.67
Repeat share 30%	53.6 ± 6.7	98.5 ± 0.6	28.3	0.63
Repeat share 50%	66.9 ± 5.9	98.6 ± 0.5	47.7	0.45
Online quality controller, $\rho = 0.5\%$	51.5 ± 6.1	98.5 ± 0.7	28.5	0.60
Online quality controller, $\rho = 2\%$	48.8 ± 4.9	98.5 ± 0.7	28.5	0.60
Online quality controller, $\rho = 5\%$	46.8 ± 4.8	98.5 ± 0.7	28.5	0.60
Online quality controller, $\eta = 0.5$	52.6 ± 8.5	97.6 ± 1.3	28.5	0.60

A wrong cache hit almost always returns a wrong answer, so quality tracks the false-hit rate closely: moving τ_{cache} from 0.80 to 0.95 raises quality retained from 97.6% to 99.0% for 2.5 points of saving. The saving grows from 33.4% with no repeats to 66.9% when half the requests are repeats. With the online quality controller, a higher audit rate only adds cost. The calibrated values were $\tau_{\text{cache}} = 0.89$, $\alpha = 0.055$ and $\lambda = 23.2$ on average, giving acceptance thresholds of 0.66 for the small model and 0.45 for the mid model. Overhead per request is 4.4 ms for the embedding and under 0.01 ms for the routing decision; a verifier call takes 3.7 s on average.

8. Discussion, Limitations and Ethics

FRUGAL-AI's savings depend on how much quality the operator is willing to trade and on how repetitive the traffic is; at the strictest targets a plain router is as good and easier to calibrate.

When it helps most. The combined pipeline pays off when a loss of 1.5 to 3 points of quality is acceptable, when traffic has many repeats, and when latency is not critical, since the verifier adds seconds to the slowest requests. When the target is within 1 point of the strongest model, or when each task type must meet the target on its own, a calibrated router is the safer choice. Three practical lessons follow. Tune against a margin, not the bare target, and report how often the target is met. Give the cache an explicit share of the error budget. Do not expect a small audit sample to steer quality online; recalibrate offline when traffic shifts.

Limitations.

- **Router training cost:** labelling requires running every pool model on a training set once. This one-time cost should be reported and amortized over expected traffic.
- **Quality measurement:** LLM judges can be biased toward longer answers or their own model family. We use task metrics where possible; we did not collect human ratings.
- **Cache risks:** wrong reuse of an answer to a similar but different question can harm users. Time-to-live, context hashing and a calibrated threshold reduce but do not remove this risk.
- **Model churn:** prices and models change often, so the router needs retraining. Recalibration needs fresh outputs from every pool model.
- **Theory assumptions:** we give no formal guarantee. The calibration margin is a heuristic, and the online-control results we cite assume i.i.d. requests and exact feedback.

Limitations of this evaluation.

- Target not always met: with the margin, FRUGAL-AI meets the 98% target in 7 of 10 seeds on the replay stream and in 5 of 10 on distinct queries. The margin makes meeting the target likely, not certain.
- Tuning rule chosen after a first evaluation: see the disclosure in Section 6.5. A fresh test set would be needed to rule out any benefit from that choice.
- Single model family: all three models are Claude models, and the large model also judges the summaries and dialogues, including its own.
- Seeds are not independent model samples: each query was answered once per model, and seeds redraw only the split, the stream and the audits.
- Small test sets: about 440 queries per split, and 40 to 200 per workload.
- Synthetic traffic: the 30% repeat share and the drift scenario are assumptions, not measurements of real traffic.
- Baselines: the 2026 systems named in Section 2 (ParetoBandit, WISERouter, Cascade Routing, Conformal Cascade) were not implemented.
- Cost method: costs are tokens times list prices from command-line calls with a measured overhead subtracted, not metered API bills.
- Possible contamination: the Codeforces problems date from 2021 and earlier, so the models may have seen them in training.

Ethics and fairness. Routing by predicted difficulty could give some user groups, languages or dialects systematically weaker models. We recommend reporting quality per language or user segment and adding per-segment quality constraints where needed. Cache entries must not leak one user's private data to another, so caches should be partitioned by tenant and exclude personal data.

Environmental impact. Fewer calls to large models also cut energy use. Reporting estimated kWh alongside cost supports Green AI reporting practice [18].

9. Conclusion and Future Work

Calibrating a semantic cache, a difficulty router and a verifier-gated cascade together, against one quality target with a margin, cut LLM inference cost by 50.8% at 98.5% of the strongest model's quality on a stream with 30% repeated requests. That is about twice the saving of routing or caching alone, and the advantage over routing holds at matched quality up to 98.5% quality retained. At a 99% target the combined pipeline is no better than a router. Prompt compression, an audit-driven online quality controller and per-workload floors for the full pipeline did not pay off, while an online budget multiplier held spend under a traffic shift at a measurable cost in quality.

Future work includes four directions:

- Contextual-bandit routers that learn from live feedback without full offline labelling.
- Joint routing over models and decoding settings, such as quantized variants and speculative decoding.
- Per-segment fairness constraints built into the controller.
- Extension to multi-step agent workflows, where one task triggers many model and tool calls.

Declarations

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Use of AI tools. The research idea, the research questions, the choice of methods and benchmarks, and all decisions during the study are the author's. The author used an AI assistant, Claude Code (Anthropic), as a tool under the author's direction to implement the experiment and analysis code, to run the analyses, and to draft and edit the manuscript text, tables and figures. The author reviewed and revised this material, checked the results against the released code and logs, and takes full responsibility for the content of this paper. The models evaluated in the experiments are Claude Haiku 5.5, Claude Sonnet 5.5 and Claude Opus 5.5; the author has no affiliation with their provider.

Code and Data Availability

The code, the logged model outputs and the results are available at https://github.com/vijayakumargr/frugal-ai. The repository rebuilds the query set from the public benchmarks and reproduces every table in this paper without model calls. It also contains the first evaluation described in Section 6.5. Version 1.0.0 is archived on Zenodo, https://doi.org/10.5281/zenodo.24202801.

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Authors, titles, venues and years were checked on 13 November 2026 against arXiv, Crossref and the proceedings pages. Three entries were corrected: the title and venue of [3] and the author lists of [4] and [7]. Entries [30] to [36] are preprints and may have been published since; check them again, and format the list to the journal's style, before submission.

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